User Manual
CodeClash Arena

Group Number: XX

Member Names and IDs

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Contents

1 Introduction

1.1 Purpose of the Manual

This user manual serves as a comprehensive guide for users of CodeClash Arena, a gamified competitive programming platform designed to transform traditional coding practice into an interactive and engaging experience. The manual provides detailed instructions on how to use the platform effectively, from account creation to participating in competitive battles and clan wars.

This document is intended for:

- Beginner programmers seeking to improve their algorithmic skills through competitive practice
- CSE students looking for engaging coding challenges
- Competitive programming enthusiasts wanting real-time head-to-head battles
- Coding instructors and educators implementing gamified learning environments
- Clan leaders organizing team-based coding competitions

The manual covers all essential features of the platform, including step-by-step instructions, visual guidance through screenshots, troubleshooting tips, and best practices for optimal user experience. Whether you are a first-time user or an experienced programmer, this guide will help you navigate the platform efficiently and make the most of its features.

1.2 Brief Overview of the Web Application

CodeClash Arena is a web-based competitive programming platform that combines the excitement of multiplayer gaming with professional coding contests. Inspired by popular competitive games like Clash Royale and Clash of Clans, and integrated with the Codeforces problem repository, the platform offers a unique blend of entertainment and skill development.

1.2.1 Core Concept

Unlike traditional coding platforms that focus solely on individual practice, CodeClash Arena introduces real-time competitive elements including:

- **1v1 Coding Battles**: Challenge other coders in head-to-head real-time competitions
- **Clan System**: Form or join teams and participate in clan-based battles

- **Rating System**: Track your progress through an ELO-based ranking mechanism
- **Practice Gym**: Solo practice mode with access to thousands of programming problems
- **Matchmaking**: Automatic pairing with opponents of similar skill levels

1.2.2 Key Differentiators

CodeClash Arena addresses limitations of conventional coding platforms by providing:

- Direct head-to-head competition with other coders in real-time
- Clan-based team competitions fostering collaboration and community
- Personalized contest creation with custom difficulty and time settings
- Gamified progression system with levels, XP, and visual achievements
- Real-time battle synchronization showing opponent progress
- AI-assisted learning through logical hints during practice sessions

1.2.3 Technical Foundation

The platform is built using modern web technologies:

- **Frontend**: React 19.2.0 with Vite 7.2.4 for fast, responsive user interface
- **Backend**: Node.js with Express 4.18.2 for server-side logic
- **Database**: Supabase (PostgreSQL) for data management
- **Problem Integration**: Codeforces API for accessing thousands of programming problems
- **Code Execution**: Piston API for local testing and Codeforces judge for official verdicts

1.2.4 Platform Features Summary

Feature	Description
Global Battle Mode	Join a matchmaking queue and compete against players of similar rating in real-time 1v1 battles

Feature	Description
Local Battle Mode	Challenge specific friends directly with customizable problem difficulty and time limits
Clan Wars	Form teams, compete in clan-based tournaments, and track team performance
Practice Gym	Access solo practice mode with filtering by difficulty, tags, and rating range
Rating System	ELO-based skill rating that adapts based on opponent strength and battle outcomes
Progression System	Earn experience points and level up through battles and problem solving
Leaderboards	Global and clan rankings updated in real-time showing top performers
Battle History	Comprehensive records of past matches, opponents, and performance metrics
Code Editor	Integrated syntax-highlighted editor supporting C++, Java, Python, JavaScript, and more
AI Hints	Logical approach guidance during practice sessions without revealing solutions
Social Features	Friend lists, clan chat, mail system, and player profile viewing

2 System Requirements

This section outlines the technical prerequisites and environmental requirements necessary to access and use CodeClash Arena effectively. Ensuring your system meets these specifications will provide optimal performance and user experience.

2.1 Supported Browsers and Devices

2.1.1 Web Browsers

CodeClash Arena is a web-based application accessible through modern web browsers. For the best experience, we recommend using the latest stable versions of the following browsers:

Browser	Minimum Version	Recommended Features
Google Chrome	Version 84+	Best performance with V8 JavaScript engine, full CSS Grid and Flexbox gap support, excellent developer tools
Mozilla Firefox	Version 103+	Strong privacy features, backdrop-filter support, reliable WebSocket connections, modern CSS features
Microsoft Edge	Version 84+	Chromium-based engine, native Windows integration, excellent performance, full modern CSS support
Safari	Version 15+	Optimized for macOS/iOS devices, complete modern CSS/JS feature support including aspect-ratio and flex gap
Opera	Version 84+	Chromium-based, built-in VPN support, efficient resource usage, modern web standards compliance
Brave	Version 1.42+	Privacy-focused, Chromium-based, full ES2020+ support, modern CSS Grid and Flexbox features

Important Browser Requirements:

- **JavaScript**: Must be enabled (required for React 19.2.0 and ES2020+ features)
- **Cookies**: Must be enabled for session management and authentication
- **Local Storage**: Browser must support HTML5 local storage (for user preferences and cached data)
- **WebSocket Support**: Required for real-time battle synchronization
- **Iframe Support**: Necessary for displaying Codeforces problem statements
- **Modern CSS Support**: CSS Grid, Flexbox with gap property, CSS custom properties (variables)
- **Advanced CSS Features**: backdrop-filter (with -webkit- prefix for Safari), CSS transforms and transitions
- **Modern JavaScript**: ES2020+ features including optional chaining (?.), nullish coalescing (??), async/await

- **Pop-up Blocker**: Should allow pop-ups from CodeClash Arena domain for certain features

Not Recommended or Not Supported:

- Internet Explorer (any version) - not supported, lacks modern JavaScript ES6+ features, CSS Grid, and CSS variables
- Safari versions below 15 - missing critical features like aspect-ratio, full flex gap support, and modern JavaScript operators
- Firefox versions below 103 - missing backdrop-filter support resulting in visual degradation
- Chrome/Edge versions below 84 - missing flex gap property causing layout issues
- Outdated browser versions - may cause compatibility issues, security vulnerabilities, and degraded performance
- Browsers with JavaScript disabled - critical functionality will not work

2.1.2 Device Compatibility

CodeClash Arena features a responsive design that adapts to various screen sizes and device types:

Device Type	**Minimum OS/Browser**	**User Experience Notes**
Desktop Computer	Windows 10+, macOS 11+, Linux	**Optimal Experience** - Full feature access (1920x1080+ recommended), dual-pane code editor, comfortable coding environment, all modern CSS/JS features supported, best for competitive battles
Laptop	Windows 10+, macOS 11+, Linux	**Excellent Experience** - All features accessible (1366x768 minimum), suitable for coding and battles, portable competitive environment, modern browser support required

Device Type	Minimum OS/Browser	User Experience Notes
iPad	iOS 15+ with Safari 15+	**Good Experience** - 10-inch or larger recommended, touch-optimized interface, suitable for reading problems and light coding, external keyboard strongly recommended for battles, full modern CSS support
Android Tablet	Android 10.0+ with Chrome 84+	**Good Experience** - 10-inch or larger recommended, touch-optimized interface, suitable for practice mode, external keyboard recommended, may require Chrome browser installation
iPhone	iOS 15+ with Safari 15+	**Limited Experience** - Best for viewing profiles, leaderboards, and battle history; coding on mobile not recommended for competitive battles; requires modern Safari for full feature support
Android Phone	Android 10.0+ with Chrome 84+	**Limited Experience** - Best for browsing and monitoring; on-screen keyboard not suitable for competitive coding; install Chrome browser for best compatibility

Important Device Notes:

- **Older Tablets**: iPads running iOS 14 or Android tablets below version 10.0 may experience visual issues (missing backdrop blur effects, layout problems with flex gap)
- **Browser Requirements**: Tablets must use Chrome 84+ (Android) or Safari 15+ (iOS) for full functionality
- **Tablet Mode Windows**: Surface devices and 2-in-1 laptops work best in desktop mode with physical keyboard attached
- **Mobile Limitations**: While responsive, mobile phones are not recommended for coding battles due to screen size constraints and on-screen keyboard limitations
- **Older Devices**: Devices more than 5 years old may struggle with modern JavaScript execution and CSS rendering

Screen Resolution Recommendations:

- **Minimum**: 1024x768 pixels (basic functionality, some layout compromises on smaller screens)
- **Recommended**: 1920x1080 pixels or higher (optimal experience with full CSS Grid layouts)
- **Competitive Battle**: 1440x900 pixels minimum (comfortable split-screen view for problem and editor)
- **Tablets**: 1280x800 or higher for iPad/Android tablets (portrait mode not recommended for battles)
- **Mobile**: 375x667 minimum for basic browsing (coding features disabled on small screens)

Input Device Recommendations:

- **Physical Keyboard**: Mandatory for competitive battles, required for efficient code input during timed contests
- **Mouse/Trackpad**: Strongly recommended for navigation, code selection, and editor interaction
- **Touch Screen**: Supported for browsing and navigation but not optimal for code editing; on-screen keyboards significantly slow down coding speed
- **Bluetooth Keyboard**: Acceptable for tablets, must support standard keyboard shortcuts (Ctrl+C, Ctrl+V, Tab, etc.)
- **External Monitor**: Optional but beneficial for multi-screen setup (problem viewing on one screen, coding on another)

Tablet-Specific Requirements:

- **iPad Users**: Requires iOS 15+ with Safari 15+, external keyboard mandatory for battles, landscape orientation recommended
- **Android Tablet Users**: Requires Android 10.0+ with Chrome 84+ installed, external keyboard strongly recommended
- **Stylus Support**: Not utilized by the platform, keyboard input is primary interaction method

2.2 Internet Connection Requirements

A stable internet connection is essential for CodeClash Arena as it operates entirely online and requires continuous connectivity for real-time features.

2.2.1 Minimum Internet Speed

Activity	**Minimum Speed**	**Description**
General Browsing	1 Mbps	Basic navigation, viewing profiles, accessing leaderboards
Practice Mode	2 Mbps	Loading problems, submitting code, receiving verdicts
1v1 Battles	5 Mbps	Real-time synchronization, live updates, timer synchronization
Clan Wars	5-10 Mbps	Multiple participants, real-time team coordination
Recommended	**10 Mbps+**	Optimal performance for all features, minimal latency

2.2.2 Connection Stability

- **Latency**: Less than 200ms ping for optimal real-time battle experience
- **Packet Loss**: Less than 1% for stable WebSocket connections
- **Connection Type**:
  - **Preferred**: Wired Ethernet connection (most stable)
  - **Acceptable**: Wi-Fi with strong signal strength (3+ bars)
  - **Not Recommended**: Mobile data/cellular (may cause disconnections during battles)

2.2.3 Critical Connectivity Features

- **Real-Time Synchronization**: During battles, the system uses WebSocket connections to synchronize:
  - Opponent submission status
  - Battle timer countdown
  - Live score updates
  - Problem acceptance notifications
- **External Service Access**: Your network must allow connections to:
  - Supabase servers (database and authentication)

  - Codeforces API endpoints (problem fetching and submission)
  - Piston API (code execution for practice mode)
  - Vercel hosting (frontend delivery)

- **Firewall Considerations**:
  - WebSocket connections (WSS protocol) must not be blocked
  - HTTPS (port 443) must be accessible
  - RESTful API calls must be permitted

Important Note: Unstable or slow internet connections may result in:

- Battle disconnections (automatic forfeit if reconnection fails)
- Delayed submission verdicts
- Real-time updates lag
- Timeout errors during code submission

2.3 Required Accounts and Permissions

2.3.1 CodeClash Arena Account

Required for: Access to all platform features

- **Registration Information Needed**:
  - Valid email address (for account verification and password recovery)
  - Username (unique identifier, 3-20 characters, alphanumeric)
  - Strong password (minimum 8 characters, recommended: uppercase, lowercase, numbers, special characters)
- **Account Verification**:
  - Email verification required upon registration
  - Verification link expires in 24 hours
  - Unverified accounts have limited functionality
- **Account Permissions**:
  - Read access to public leaderboards and profiles
  - Write access to personal profile information
  - Participation rights in battles, practice sessions, and clan activities
  - Submission rights for code solutions

2.3.2 Codeforces Account

Required for: Code submission and official verdict retrieval
Requirement Level: Mandatory for competitive battles and official problem solving

- **Why Codeforces Account is Needed**:
  - CodeClash Arena integrates with Codeforces for problem sourcing
  - All code submissions are forwarded to Codeforces judge for verification
  - Official Codeforces verdicts (Accepted, Wrong Answer, TLE, etc.) are retrieved
  - Problems are displayed via iframe from Codeforces platform
- **Creating a Codeforces Account**:
  - Visit `https://codeforces.com/register`
  - Complete registration with email verification
  - Choose a unique handle (username)
  - Activate account through confirmation email
- **Linking Codeforces Account to CodeClash Arena**:
  - Navigate to Profile Settings in CodeClash Arena
  - Enter your Codeforces handle in the designated field
  - System validates the handle existence
  - Handle is stored and used for all future submissions
- **Account Permissions Required**:
  - Active Codeforces account (not banned or restricted)
  - Ability to submit solutions to Codeforces problems
  - Access to Codeforces problem archive

Important Notes:

- Without a linked Codeforces account, users cannot:
  - Submit code during battles (automatic forfeit)
  - Participate in competitive ranked matches
  - Receive official verdict status
  - Contribute to clan battle scores

- Users can still access:
  - Platform browsing and exploration
  - Leaderboard viewing
  - Profile viewing
  - Problem browsing (without submission)
- Codeforces handle cannot be changed frequently (update limit implemented to prevent abuse)
- Ensure your Codeforces account is in good standing (not restricted for rule violations)

2.3.3 Additional Permissions

- **Browser Permissions**:
  - Cookies: Required for session management and authentication persistence
  - Local Storage: Required for caching user preferences and temporary data
  - Clipboard Access: Optional, improves code copying experience
- **Third-Party Service Access**:
  - Supabase (authentication and database): No additional account needed
  - Piston API (code execution): No additional account needed
  - Vercel hosting: No user account required

2.4 Software Requirements

2.4.1 Operating System Compatibility

CodeClash Arena is a cross-platform web application compatible with any operating system that supports modern web browsers:

- **Windows**: Windows 10 or later (Windows 11 recommended)
- **macOS**: macOS 11 (Big Sur) or later (required for Safari 15+)
- **Linux**: Any modern distribution (Ubuntu 20.04+, Fedora 33+, etc.)
- **Chrome OS**: Latest stable version (Chrome 84+ equivalent)
- **iOS**: iOS 15 or later (required for Safari 15+ with full feature support)
- **Android**: Android 10.0 or later (for Chrome 84+ support)

2.4.2 No Installation Required

- CodeClash Arena is entirely browser-based
- No desktop application or mobile app download necessary
- No system administrative privileges required
- Access the platform directly at: `[your-deployment-url]`

2.4.3 Optional Enhancements

- **Code Editor Extensions**: While the platform provides an integrated code editor, advanced users may prefer:
  - Copying code to external IDEs (VS Code, Sublime Text) for complex solutions
  - Using browser extensions for enhanced syntax highlighting
- **Screen Management**: For optimal battle experience:
  - Multiple monitor setup (problem on one screen, editor on another)
  - Split-screen window management
  - Browser tab management for simultaneous problem viewing

2.5 Hardware Recommendations

Component	Minimum	Recommended	Purpose
Processor	Dual-core 2.0 GHz	Quad-core 2.5 GHz+	Fast page rendering, smooth real-time updates
RAM	4 GB	8 GB or more	Multiple browser tabs, smooth multitasking
Storage	N/A (web-based)	N/A	No local installation required
Display	1024x768 resolution	1920x1080 or higher	Comfortable coding environment, split-screen view

2.6 Network Security Considerations

- **Corporate Networks**: Some corporate firewalls may block:
  - WebSocket connections (required for real-time battles)

  - External API calls (Codeforces, Piston API)
  - Iframe embedding (problem display)

  Solution: Contact your network administrator to whitelist necessary domains

- **VPN Users**: VPN connections may introduce latency affecting real-time battle performance
- **Proxy Servers**: Ensure proxy configuration allows WebSocket (WSS) connections

2.7 Accessibility Requirements

- **Visual**: Minimum 768px width viewport for usable interface
- **Color Scheme**: System adapts to different lighting conditions; custom themes may be implemented in future versions
- **Font Rendering**: Browser must support standard web fonts for proper code editor display

2.8 Modern Web Features and Browser Compatibility

CodeClash Arena utilizes cutting-edge web technologies that require modern browser support. The following features are implemented:

- **CSS Features**: Grid Layout, Flexbox with gap property, CSS Custom Properties (variables), backdrop-filter for blur effects, transforms and transitions
- **JavaScript Features**: ES2020+ syntax including optional chaining (?.), nullish coalescing (??), async/await, arrow functions, destructuring, template literals
- **Web APIs**: WebSocket (real-time communication), Fetch API (HTTP requests), Local Storage API, History API (routing), Clipboard API
- **React 19.2.0**: Modern Hooks (useState, useEffect, useContext), latest rendering optimizations

Minimum Browser Versions for Full Compatibility:

- Chrome/Edge 84+ (for Flexbox gap support)
- Firefox 103+ (for backdrop-filter support)
- Safari 15+ (for complete modern CSS/JS features)
- Mobile Safari iOS 15+ (for full mobile compatibility)

2.9 System Requirements Summary

Requirement Category	Status
Modern Web Browser (Chrome 84+, Firefox 103+, Edge 84+, Safari 15+)	MANDATORY
JavaScript Enabled (ES2020+ support)	MANDATORY
Modern CSS Support (Grid, Flexbox gap, backdrop-filter)	MANDATORY
Internet Connection (5+ Mbps recommended)	MANDATORY
CodeClash Arena Account	MANDATORY
Codeforces Account (linked handle)	MANDATORY
Desktop/Laptop Device (for competitive battles)	MANDATORY
Physical Keyboard (for competitive battles)	MANDATORY
Stable Internet (Wired connection preferred)	**RECOMMENDED**

Note: Mobile devices (tablets, phones) can be used for browsing profiles, viewing leaderboards, and monitoring battles, but Desktop/Laptop with Physical Keyboard is mandatory for participating in competitive coding battles.

3 Getting Started

This section provides step-by-step instructions for new users to access CodeClash Arena, create an account, log in, and navigate the initial setup process.

3.1 Accessing the Platform

CodeClash Arena is accessible through any modern web browser without requiring software installation.

Steps to Access:

1. Open your preferred web browser (Chrome 84+, Firefox 103+, Edge 84+, or Safari 15+)
2. Navigate to the CodeClash Arena URL: [`https://code-clash-arena-six.vercel.app/`]
3. Ensure JavaScript is enabled in your browser settings
4. The homepage will load, displaying the main landing page

First-Time Visitors: Upon accessing the platform for the first time, you will see the homepage featuring:

- Main title: "HELLO CODERS, WELCOME!"
- **GET STARTED** button - Click to proceed to the Login page
- **About us** link in the header - Learn about the platform
- **Learn More** link in the header - Discover features and capabilities
- Animated particle effects for visual engagement

3.2 Sign-Up Process

New users must create a CodeClash Arena account to access platform features. The sign-up process is straightforward and requires basic information.

3.2.1 Sign-Up Steps

Step 1: Navigate to Sign-Up Page

- From the homepage, click the **"GET STARTED"** button
- You will be redirected to the Login page first
- On the Login page, locate the text at the bottom: **"Don't have an account?"**
- Click the **"Sign Up"** link (highlighted in gold/yellow color)
- The Sign-Up page will load with registration form

Step 2: Enter Registration Information
Fill in all three required fields in the sign-up form:

- **Email**: Provide a valid email address
  - Used for account verification and password recovery
  - Must be unique (not already registered in the system)
  - Ensure you have access to this email for verification
  - Format: username@domain.com
- **Codeforces Handle**: Enter your existing Codeforces username
  - **This is your Codeforces account username** - not a new username
  - Must be a valid, existing Codeforces handle

homepage.png

Figure 1: CodeClash Arena Loginpage - Locate Sign Up button

  - This handle is linked immediately during registration
  - Used for all code submissions and battle participation
  - If you don't have a Codeforces account, create one first at `https://codeforces.com/register`

- **Password**: Create a strong password for your CodeClash Arena account
  - Minimum 8 characters required
  - Recommended: Include uppercase, lowercase, numbers, and special characters
  - Password is case-sensitive
  - This is separate from your Codeforces password

Important Note: Unlike typical registrations, CodeClash Arena requires your Codeforces handle during sign-up because the platform integrates directly with Codeforces for problem sourcing and code submission.

signup_form.png

Figure 2: Sign-Up Form - Enter registration details

Step 3: Submit Registration

- Click the **"SignUp"** button (green/teal colored button)
- The system will validate all entered information
- If validation fails, field labels will turn red indicating errors:
  - Red label on Email - Check email format or email may already be registered
  - Red label on Codeforces Handle - Verify handle exists on Codeforces
  - Red label on Password - Ensure password meets minimum requirements (8+ characters)

  - Empty fields are highlighted in red when attempting to submit

- Upon successful validation, account creation begins

signup_submit.png

Figure 3: Sign-Up Form Completion - Click SignUp button

Step 4: Email Verification Alert

- After successful registration, a success alert popup appears with the message: **"Signup successful! Welcome, check your email [your-email@domain.com]"**
- The page automatically redirects to the Homepage after 1.5 seconds
- Check your email inbox for a verification message from Supabase (CodeClash Arena's authentication provider)
- Subject line typically: "Confirm Your Email" or "Verify Your Email"
- Email may take 1-5 minutes to arrive

- Check spam/junk folder if email is not received within 5 minutes

verification_email.png

Figure 4: Email Verification

Step 5: Verify Email Address

- Open the verification email
- Click on the **"Verify Email Address"** button or link
- Browser will open and redirect to CodeClash Arena
- Verification confirmation page appears
- Account is now fully activated

Step 6: Login to Complete Setup

After email verification:

- Navigate to the Login page
- Login using your registered email and password
- Your Codeforces handle is already linked from registration
- You can now access all platform features including battles, practice gym, and clan wars

Important Notes:

- **Codeforces Handle Required First**: You must have an existing Codeforces account before signing up on CodeClash Arena
- **Verification Link Expiry**: Email verification links typically expire after 24 hours
- **Unverified Accounts**: You must verify your email to login and access platform features
- **Handle Validation**: The system stores your Codeforces handle but doesn't validate its existence during registration - ensure you enter it correctly
- **Account Security**: Never share your CodeClash Arena password with anyone (keep it separate from your Codeforces password)
- **Success Alert**: Green alert popup confirms successful registration
- **Error Alert**: Red alert popup displays error messages if registration fails

3.3 Login Process

Existing users can access their accounts through the login page. The process is quick and secure.

3.3.1 Login Steps

Step 1: Navigate to Login Page

- From the homepage, click the **"GET STARTED"** button
- You will be automatically redirected to the Login page
- Alternatively, navigate directly via your browser if you know the URL
- The login page displays with the CodeClash Arena logo at the top
- A **"← Back"** button is available in the top-left corner to return to the homepage

Step 2: Enter Login Credentials

loginpage.png

Figure 5: Loginpage - Navigate to Login

- **Email Field**: Enter the email address used during registration
  - Labeled as "Email" with a user icon on the left
  - Email is case-insensitive
  - Ensure no extra spaces before or after email
  - Floating label design - label moves up when you start typing
- **Password Field**: Enter your account password
  - Labeled as "Password" with floating label design
  - Password is case-sensitive
  - Click on the password field to focus it

- When focused, an **eye icon** appears on the right side
- Click the eye icon to toggle password visibility (show/hide)
- Closed eye icon = password hidden; Open eye icon = password visible

Step 3: Submit Login

- Click the **"Log In"** button (golden/yellow colored button with login icon)
- System validates credentials with Supabase authentication service
- Authentication typically completes within 1-2 seconds
- Success or error alert popup will appear

`login_submit.png`

Figure 6: Login successful - Success alert displayed

Step 4: Successful Login - Redirection

- Upon successful authentication, a green success alert appears: **"Sign in successful!"**
- After 1.5 seconds, you are automatically redirected to the **Main Dashboard** (/main)
- Your Codeforces handle is automatically loaded from the database and stored in browser localStorage
- Navigation bar updates to show logged-in status with your profile information
- Access to all platform features is now available: PvP Battles, Practice Gym, Clan Wars, Leaderboards

maindasboard.png

Figure 7: Main Dashboard - After successful login

Common Login Errors and Solutions:

Error Message	Solution
"Invalid email or password" or authentication error	Verify email and password are correct; check for typos; ensure Caps Lock is off; verify email has been verified via link sent during signup
"Email not verified"	Check your email inbox for verification link; you must verify email before logging in; check spam/junk folder
"Account not found"	Email may not be registered; try signing up or verify correct email address
"Too many login attempts"	Wait 15 minutes before trying again; security measure against brute force attacks
"Network error"	Check internet connection; ensure stable connectivity; try refreshing page

3.3.2 First-Time Login - Account Setup Complete

After your first successful login:

Codeforces Account Already Linked:

- Your Codeforces handle was linked during the sign-up process
- No additional Codeforces setup is required
- The system automatically retrieves your cf_handle from the database upon login
- This handle is used for all code submissions to Codeforces during battles
- You can proceed immediately to explore the platform features

If You Need to Update Codeforces Handle:

1. Navigate to **Profile Settings** or **User Profile** section
2. Locate the **"Codeforces Handle"** field
3. Update your Codeforces username/handle if needed
4. Save changes
5. Note: Handle changes may be rate-limited to prevent abuse

Important: Without a linked Codeforces account, you cannot participate in competitive battles or submit code solutions.

3.4 Logout Process

Users can securely log out of their accounts at any time to end their session.

3.4.1 Logout Steps

Step 1: Access User Menu

- Locate your username or profile icon in the top-right corner of the navigation bar
- Click on the username/profile icon to open the dropdown menu
- User menu will display with various options

logout.png

Figure 8: User Menu - Access logout option

Step 2: Select Logout Option

- From the dropdown menu, locate and click **"Logout"** or **"Sign Out"**

- Option is typically at the bottom of the menu
- May be represented by a logout icon (exit door or arrow)

Step 3: Confirm Logout (if applicable)

- Confirmation dialog may appear: "Are you sure you want to logout?"
- Click **"Yes"** or **"Confirm"** to proceed
- Click **"Cancel"** to remain logged in
- Some implementations may log out immediately without confirmation

Step 4: Logout Completion

- Session is terminated on the server
- Authentication tokens are cleared from browser
- You are redirected to the homepage or login page
- Confirmation message appears: "Successfully logged out"
- All session data is cleared for security

Important Security Notes:

- **Always logout on shared computers**: Prevent unauthorized access to your account
- **Battle interruption**: If logged out during an active battle, it counts as forfeit
- **Unsaved changes**: Any unsaved draft code in editor may be lost
- **Session persistence**: "Remember Me" option keeps you logged in until explicit logout
- **Automatic logout**: Sessions may expire after extended inactivity (typically 24-48 hours)

3.5 Troubleshooting Common Issues

3.5.1 Cannot Receive Verification/Reset Emails

Solutions:

- Check spam/junk folder for emails from Supabase
- Verify email address was entered correctly during signup
- Wait 5-10 minutes (email delivery may be delayed)

- Whitelist emails from your authentication provider
- Check email provider's filtering settings
- Contact platform support if verification email is never received

3.5.2 Account Locked After Multiple Failed Login Attempts

Solutions:

- Wait 15-30 minutes before attempting to login again
- Ensure correct email and password are being used
- Use password reset if password is forgotten
- Contact support if issue persists after waiting period

3.5.3 Browser Compatibility Issues

Solutions:

- Update browser to minimum required version (Chrome 84+, Firefox 103+, Safari 15+)
- Enable JavaScript in browser settings
- Clear browser cache and cookies
- Disable browser extensions that may interfere
- Try using a different supported browser

3.6 Next Steps After Getting Started

Once successfully logged in, proceed to:

1. **Explore Main Dashboard**: Familiarize yourself with navigation menu and available features - PvP Battles, Practice Gym, Clan Wars, and more
2. **Verify Codeforces Handle**: Your Codeforces handle was linked during signup - verify it appears correctly in your profile
3. **Complete Profile**: Add additional information, avatar, or bio (if available in profile settings)
4. **Browse Practice Gym**: Start with practice problems to familiarize yourself with the code editor and problem-solving interface

5. **Join or Create a Clan**: Team up with other coders for clan battles and collaborative competitions (optional)
6. **Enter First Battle**: Once comfortable, try a 1v1 PvP battle to experience competitive real-time coding

Recommended First Actions:

- Visit the **Practice Gym** to try solving a problem without time pressure
- Check the **Leaderboard** to see top-ranked players
- Read platform tutorials or help section (if available)
- Configure notification preferences in settings

4 User Roles

CodeClash Arena is designed exclusively for authenticated registered users to ensure fair participation in competitive coding battles. Access to the platform requires account creation, email verification, and a linked Codeforces handle. There are no guest or admin roles in the system - all users have equal access to platform features and compete on a level playing field based on their coding skills and performance.

4.1 Registered User

A registered user is any individual who has successfully completed the sign-up process, verified their email address, and linked a valid Codeforces handle to their account. This is the only role on the platform, and all authenticated users operate with the same base permissions.

4.1.1 Permissions

Registered users have access to the following platform features:

- **Account Management**: Register an account with email and Codeforces handle, log in and log out securely, update profile information and preferences
- **Real-Time 1v1 Coding Battles**: Participate in Local battles (challenge specific friends with custom settings), participate in Global battles (matchmaking with players of similar skill level), compete in time-based coding challenges with live synchronization
- **Clan System**: Create a new clan or join existing clans, participate in clan battles when selected by clan leader, contribute to clan rankings and statistics, manage clan memberships (leaders can approve/reject join requests)

- **Code Submission**: Submit code solutions in multiple programming languages (C++, Java, Python, JavaScript, etc.), receive automated verdicts from Codeforces judge (Accepted, Wrong Answer, Time Limit Exceeded, etc.), test code locally using integrated code editor
- **Statistics and Rankings**: View personal battle history and performance metrics, track XP, level, and ELO-based rating progression, access global and clan leaderboards, view detailed submission verdicts and problem-solving statistics
- **Practice Mode**: Access Practice Gym with thousands of Codeforces problems, filter problems by difficulty, tags, and rating range, receive AI-based logical hints for approach guidance (without revealing full solutions), solve problems without time pressure for skill improvement
- **Social Features**: Send and accept friend requests, view friend profiles and statistics, receive notifications for clan join requests and battle invitations

4.1.2 Restrictions

Registered users are subject to the following restrictions to maintain platform integrity and fair competition:

- **No System Modifications**: Users cannot modify system configurations, settings, or platform code
- **No Ranking Manipulation**: Direct manipulation of ratings, XP, leaderboards, or battle outcomes is strictly prohibited
- **No Private Data Access**: Users cannot access other users' private information, submission code, or system database records
- **No Unauthorized Authentication**: Users cannot share accounts or use multiple accounts to gain unfair advantages
- **Fair Play Requirements**: Code submissions must be original work; plagiarism or copying solutions is not permitted

5 Features and Functionality

This section provides detailed explanations of all major features available in CodeClash Arena, including step-by-step usage instructions, input/output descriptions, and visual guidance through UI screenshots.

5.1 PvP Battle System

The Player vs Player (PvP) battle system is the core competitive feature of CodeClash Arena, allowing users to engage in real-time coding competitions. The system offers two distinct battle modes: Global Battle Mode for matchmaking with similarly skilled opponents, and Local Battle Mode for challenging specific friends with customizable settings.

5.1.1 Global Battle Mode

Purpose: Global Battle Mode provides an automated matchmaking system that pairs you with opponents of similar skill level for fair and competitive 1v1 coding battles. Your rating changes based on battle outcomes and opponent strength using an ELO-based ranking system.

Step-by-Step Usage:

Step 1: Access Global Battle Mode

1. From the Main Dashboard, locate the **"Attack"** button on the left navigation menu
2. Click the **"Attack"** button to open the battle mode selection overlay
3. The overlay menu displays three options: Global Battle, Local Battle, and Practice Gym
4. Click on **"Global Battle"** option (typically marked with a globe or trophy icon)
5. The Global Battle interface loads

[Screenshot: Main Dashboard with Attack button highlighted]

Figure 9: Main Dashboard - Click Attack button to access battle modes

Step 2: Enter Matchmaking Queue

1. On the Global Battle page, you will see your current rating, level, and matchmaking options
2. Click the **"Find Match"** or **"Start Matchmaking"** button
3. The system searches for an available opponent with similar rating (±200 rating range)
4. A loading/searching animation displays with message: "Searching for opponent..."
5. Queue time varies based on available players (typically 10-60 seconds)
6. You can cancel matchmaking by clicking **"Cancel Search"** button

[Screenshot: Global Battle matchmaking screen with "Find Match" button highlighted]

Figure 10: Global Battle - Enter matchmaking queue

Step 3: Match Found - Battle Initialization

1. When an opponent is found, a notification appears: "Opponent Found!"
2. Opponent's username, rating, and level are displayed
3. Battle settings are shown:
   - Problem difficulty (automatically selected based on average rating)
   - Battle duration (standard time limit, typically 30-45 minutes)
   - Programming languages allowed
4. Countdown timer begins (3-5 seconds) before battle starts
5. Both players are synchronized and enter the battle arena simultaneously

Step 4: Battle Arena Interface

[Screenshot: Match found screen showing opponent details and countdown]

Figure 11: Match Found - Opponent details and battle initialization

1. Battle arena loads with split-screen layout:
   - **Left Panel**: Problem statement (Codeforces problem embedded in iframe)
   - **Right Panel**: Code editor with syntax highlighting
   - **Top Bar**: Battle timer, your username, opponent username, submission status indicators
2. Battle timer starts counting down from the allocated time
3. Read the problem statement carefully
4. Opponent's submission status is visible in real-time (e.g., "Opponent is coding...", "Opponent submitted")

[Screenshot: Battle arena showing problem statement on left, code editor on right, timer at top]

Figure 12: Battle Arena - Split-screen interface with problem and editor

Step 5: Writing and Testing Code

1. Select your preferred programming language from the language dropdown menu (C++, Java, Python, JavaScript, etc.)
2. Write your solution code in the integrated editor
3. Use the **"Run Code"** or **"Test"** button to test your code locally:
   - Input sample test cases manually
   - View output and execution time
   - Debugging assistance through local testing (uses Piston API)
4. Fix any errors or logic issues identified during testing
5. Refine your solution for edge cases

[Screenshot: Code editor with language dropdown highlighted and Run Code button marked]

Figure 13: Code Editor - Language selection and local testing

Step 6: Submitting Solution

1. Once confident in your solution, click the **"Submit"** button
2. Code is sent to Codeforces judge for official evaluation
3. Submission status appears: "Submitting..." → "In Queue" → "Running on Test X"
4. Wait for verdict (typically 5-15 seconds, depending on Codeforces server load)
5. Verdict is displayed:
   - **Accepted (AC)**: All test cases passed - you win the battle!
   - **Wrong Answer (WA)**: Failed test case - you can resubmit

- **Time Limit Exceeded (TLE)**: Code too slow - optimize and resubmit
- **Runtime Error (RE)**: Code crashed - fix bugs and resubmit
- **Compilation Error (CE)**: Syntax errors - fix and resubmit

6. You can submit multiple times until you get Accepted or time runs out

[Screenshot: Submit button highlighted and verdict display showing "Accepted"]

Figure 14: Code Submission - Submit button and verdict display

Step 7: Battle Conclusion

1. Battle ends when:
   - One player gets Accepted verdict (winner determined)
   - Battle timer reaches zero (player with most progress wins, or draw if equal)
   - Both players get Accepted (first to submit wins)
2. Battle result screen displays:
   - Winner/Loser announcement
   - Final submission status for both players
   - Time taken by each player
   - Rating change (e.g., "+15 rating" for winner, "-12 rating" for loser)
   - XP earned (winners earn more XP)
3. Battle summary is added to Battle History
4. Click **"Return to Dashboard"** to exit battle arena

Input Requirements:

[Screenshot: Battle result screen showing winner, rating change, and XP earned]

Figure 15: Battle Result - Winner announcement and statistics

- **Player Prerequisites**: Valid CodeClash Arena account, linked Codeforces handle, stable internet connection (5+ Mbps)
- **Matchmaking Input**: Player rating (automatically used for matchmaking), availability in matchmaking pool
- **Code Input**: Solution code in selected programming language, must compile without syntax errors

Output and Results:

- **Matchmaking Output**: Opponent found with username, rating, level
- **Battle Output**: Real-time battle interface with synchronized timer, problem statement, code editor
- **Submission Output**: Official Codeforces verdict (AC, WA, TLE, RE, CE, etc.), test case results, execution time
- **Battle Conclusion Output**: Win/Loss/Draw result, rating change (ELO-based calculation), XP earned, updated statistics in profile, battle record in history

Special Features:

- **Real-Time Synchronization**: See opponent's submission status live ("Opponent submitted", "Opponent got Accepted")
- **Automatic Forfeiture**: Disconnection for ¿30 seconds results in automatic loss
- **Fair Matchmaking**: Rating-based pairing ensures competitive balance
- **Multiple Submissions**: Unlimited submission attempts during battle time

5.1.2 Local Battle Mode

Purpose: Local Battle Mode allows you to challenge specific friends or users directly, with customizable battle settings including problem difficulty, time limits, and problem selection. This mode is ideal for practice with friends or organizing custom competitions.

Step-by-Step Usage:

Step 1: Access Local Battle Mode

1. From the Main Dashboard, click the **"Attack"** button on the left navigation menu
2. The battle mode selection overlay appears
3. Click on **"Local Battle"** option (typically marked with a duel icon or crossed swords)
4. The Local Battle configuration page loads

[Screenshot: Battle mode overlay with Local Battle option highlighted]

Figure 16: Battle Mode Selection - Local Battle option

Step 2: Configure Battle Settings

1. On the Local Battle setup page, configure the following settings:
   - **Opponent Selection**:
     - Enter opponent's username/handle in the search field
     - System validates that user exists
     - Friend list may be displayed for quick selection
   - **Problem Difficulty**:
     - Select from difficulty levels: Easy (800-1200 rating), Medium (1200-1800 rating), Hard (1800-2400 rating), Expert (2400+)

  - Or specify exact Codeforces rating range
- **Time Limit**:
  - Choose battle duration: 15 minutes, 30 minutes, 45 minutes, 60 minutes, or custom
- **Problem Selection** (Optional):
  - Let system auto-select from difficulty range
  - Or manually specify a Codeforces problem by contest ID and problem letter

2. Review all settings before proceeding

[Screenshot: Local Battle configuration page with opponent field, difficulty selector, and time limit options marked]

Figure 17: Local Battle Configuration - Battle settings customization

Step 3: Send Battle Invitation

1. After configuring settings, click the **"Send Challenge"** or **"Invite to Battle"** button
2. System validates that opponent is online and available
3. Battle invitation is sent to the opponent with configured settings displayed
4. You see a waiting screen: "Waiting for [opponent name] to accept..."
5. Invitation includes:
   - Your username as challenger
   - Selected difficulty and time limit
   - Option to accept or decline

[Screenshot: Battle invitation sent screen showing waiting status]

Figure 18: Battle Invitation - Waiting for opponent acceptance

6. You can cancel the invitation while waiting by clicking **"Cancel Invitation"**

Step 4: Opponent Accepts - Battle Begins

1. When opponent accepts, both players receive notification: "Battle Starting!"
2. Countdown timer (3-5 seconds) synchronizes both players
3. Battle arena loads with the configured problem and settings
4. Interface is identical to Global Battle mode:
   - Problem statement on left
   - Code editor on right
   - Timer countdown from configured duration
   - Opponent status visible in real-time
5. Both players start coding simultaneously

Step 5-7: Battle Execution

The remaining steps (writing code, testing, submitting, and battle conclusion) follow the same process as Global Battle Mode (refer to Global Battle Mode Steps 5-7).

Key differences in Local Battle:

- Battle uses your custom time limit (not standard duration)
- Problem difficulty is as configured (not auto-selected by system)
- Rating changes may be smaller or disabled (depending on implementation)
- Results still recorded in Battle History

[Screenshot: Local Battle arena showing custom settings in header (time limit displayed)]

Figure 19: Local Battle Arena - Customized battle in progress

Accepting Incoming Local Battle Invitations:

When you receive a Local Battle invitation from another player:

1. Notification appears (popup or notification badge)
2. Click notification to view invitation details
3. Invitation shows: Challenger's name, difficulty level, time limit, problem (if specified)
4. Click **"Accept"** to enter battle, or **"Decline"** to reject
5. If accepted, battle arena loads immediately with countdown

[Screenshot: Incoming battle invitation popup with Accept and Decline buttons highlighted]

Figure 20: Incoming Battle Invitation - Accept or Decline

Input Requirements:

- **Challenger Input**: Opponent username (must exist and be online), difficulty selection, time limit, optional problem specification
- **Opponent Input**: Accept or decline decision
- **Code Input**: Solution code (same as Global Battle)

Output and Results:

- **Invitation Output**: Invitation sent/received notification, waiting status
- **Battle Output**: Custom battle arena with configured settings
- **Results Output**: Battle outcome (Win/Loss/Draw), rating change (if applicable), XP earned, battle added to history

Special Features:

- **Flexible Settings**: Complete control over difficulty and duration
- **Friend Practice**: Practice with friends without rating pressure
- **Custom Problems**: Specify exact problems for focused practice
- **Invitation System**: Organized challenge workflow with accept/decline options

5.2 Practice Gym

Purpose: Practice Gym provides a solo training environment where you can solve coding problems without time pressure or competitive stress. Access thousands of Codeforces problems, filter by difficulty and tags, receive AI-based hints, and improve your skills at your own pace.

Step-by-Step Usage:

Step 1: Access Practice Gym

1. From the Main Dashboard, click the **"Attack"** button
2. In the battle mode overlay, click **"Practice Gym"** option (typically marked with a dumbbell or training icon)
3. Practice Gym interface loads showing problem browser

Step 2: Browse and Filter Problems

1. The Practice Gym displays a list of available problems with filtering options:
    - **Difficulty Filter**: Select rating range (e.g., 800-1000, 1200-1400, etc.)

[Screenshot: Battle mode overlay with Practice Gym option highlighted]

Figure 21: Battle Mode Selection - Practice Gym option

- **Tag Filter**: Choose problem categories (e.g., Dynamic Programming, Graphs, Greedy, Math, etc.)
- **Search**: Search by problem name or contest ID
- **Solved Status**: Filter to show only unsolved problems or include solved

2. Each problem displays:
   - Problem name and ID (e.g., "1500A - Beautiful Numbers")
   - Difficulty rating (e.g., 1200)
   - Tags (e.g., "greedy, math")
   - Solved status (checkmark if previously solved)
3. Apply filters to narrow down problem selection
4. Browse through paginated problem list

Step 3: Select a Problem

1. Click on a problem from the list to open it
2. Problem page loads with:
   - Problem statement (Codeforces problem embedded in iframe or rendered text)
   - Input/output specifications
   - Sample test cases
   - Constraints and notes
3. Read the problem carefully

[Screenshot: Practice Gym problem browser with difficulty filter, tag filter, and problem list visible]

Figure 22: Practice Gym - Problem browser with filtering options

4. Analyze sample inputs and outputs
5. Click **"Start Solving"** or **"Open Editor"** to begin coding

[Screenshot: Problem statement view with "Start Solving" button highlighted]

Figure 23: Practice Gym - Problem statement display

Step 4: Solving the Problem

1. Practice solving interface loads with:
   - **Left Panel**: Problem statement (persistent view)
   - **Right Panel**: Code editor
   - **Top Bar**: Problem title, timer (optional), hint button
2. Select programming language from dropdown
3. Write your solution code

4. No time limit - work at your own pace
5. Code editor supports syntax highlighting, auto-indentation, and basic autocomplete

[Screenshot: Practice solving interface with problem on left, code editor on right]

Figure 24: Practice Gym - Solving interface

Step 5: Local Testing

1. Use the **"Run Code"** or **"Test"** button to test your solution locally
2. Input test cases:
   - Use provided sample test cases (auto-populated)
   - Or enter custom test cases manually
3. Click **"Run"** to execute code
4. View output results:
   - Expected output vs. your output
   - Execution time
   - Any runtime errors or compilation errors
5. Debug and refine your solution based on test results
6. Run multiple tests with different inputs

Step 6: AI Hint System (Optional)

1. If stuck on the problem, click the **"Get Hint"** button (lightbulb icon)
2. AI hint system provides logical guidance without revealing full solution:

[Screenshot: Local testing panel showing input, expected output, actual output, and Run button highlighted]

Figure 25: Practice Gym - Local code testing

- Hint Level 1: General approach or algorithm category (e.g., "Consider using a greedy approach")
- Hint Level 2: More specific guidance (e.g., "Sort the array first, then iterate")
- Hint Level 3: Detailed steps without code (e.g., "Maintain two pointers...")

3. Hints are progressive - start with Level 1, request more if needed
4. Each hint displays in a popup or side panel
5. Use hints to learn problem-solving strategies

[Screenshot: Hint button highlighted and hint popup showing logical guidance]

Figure 26: Practice Gym - AI hint system

Step 7: Submit Solution

1. Once confident in your solution, click **"Submit"** button

2. Code is submitted to Codeforces judge for official evaluation
3. Verdict process (same as battle mode):
   - "Submitting..." → "In Queue" → "Running on Test X"
   - Final verdict: Accepted, Wrong Answer, TLE, RE, CE, etc.
4. If Accepted:
   - Success notification: "Problem Solved!"
   - XP is awarded (less than battle mode, but still contributes to progression)
   - Problem marked as solved in your profile
   - Checkmark appears next to problem in problem list
5. If not Accepted:
   - Review verdict and failed test case (if available)
   - Modify code and resubmit
   - Unlimited submission attempts

[Screenshot: Submit button and verdict display showing "Accepted" with XP earned]

Figure 27: Practice Gym - Submission and verdict

Step 8: Return to Problem Browser

1. After solving (or at any time), click **"Back to Problems"** or **"Exit"** button
2. Return to Practice Gym problem browser
3. Solved problem now shows checkmark indicator
4. Statistics updated: total problems solved, difficulty distribution

5. Select another problem to continue practicing

Input Requirements:

- **Problem Selection**: Choose difficulty range, tags, or specific problem
- **Code Input**: Solution code in selected language
- **Test Input**: Sample or custom test cases for local testing
- **Hint Request**: Optional - request AI assistance

Output and Results:

- **Problem List Output**: Filtered problems matching criteria, difficulty ratings, tags
- **Local Test Output**: Execution results, output comparison, runtime
- **Hint Output**: AI-generated logical guidance (progressive levels)
- **Submission Output**: Official Codeforces verdict, XP earned (if Accepted), solved status update
- **Statistics Output**: Updated solve count, progress tracking

Special Features:

- **No Time Pressure**: Learn and practice at your own pace
- **AI Hints**: Intelligent guidance system for learning
- **Extensive Problem Library**: Access to thousands of Codeforces problems
- **Progress Tracking**: Monitor which problems you've solved
- **Local Testing**: Test code before submission without penalty
- **XP Rewards**: Earn experience points for skill progression

5.3 Clan System

The Clan System enables team-based competitive coding through collaborative clans. Users can create clans, recruit members, and participate in clan battles where teams compete for points and rankings.

5.3.1 Creating and Joining Clans

Purpose: Form or join coding teams to participate in clan wars, collaborate with teammates, and compete for clan rankings and glory.

Creating a New Clan:

Step 1: Access Clan Interface

1. From Main Dashboard, click **"Your Clan"** button on the left navigation menu
2. If you are not in a clan, the clan browsing menu appears
3. Interface shows options: **"Create Clan"**, **"Find Clan"**, and list of available clans

[Screenshot: Clan menu showing Create Clan and Find Clan options]

Figure 28: Clan Menu - Create or Join options for users without clan

Step 2: Initiate Clan Creation

1. Click the **"Create Clan"** button
2. Clan creation form appears with required fields

Step 3: Fill Clan Information

1. Enter clan details in the creation form:
   - **Clan Name**: Unique name for your clan (3-30 characters, alphanumeric)
   - **Clan Tag**: Short identifier (2-5 characters, appears in brackets, e.g., [TAG])
   - **Description**: Brief description of clan (optional, up to 200 characters)
   - **Clan Type**: Public (anyone can request to join) or Private (invite-only)
   - **Member Limit**: Maximum members allowed (typically 10-50)
   - **Minimum Rating**: Required rating for joining (optional filter)

[Screenshot: Clan creation form with fields for name, tag, description, type, and Create button highlighted]

Figure 29: Clan Creation - Form with all required fields

2. Review all information
3. Click **"Create Clan"** button to finalize

Step 4: Clan Created Successfully

1. Success notification: "Clan created successfully!"
2. You are automatically set as Clan Leader
3. Redirected to your clan's main page showing:
   - Clan name, tag, and description
   - Member list (currently only you)
   - Clan statistics: Total points (0), battles fought (0), wins/losses
   - Join requests section (empty initially)
4. Clan is now visible to other users in clan browsing list

Joining an Existing Clan:
Step 1: Browse Available Clans

1. From the clan menu (accessed via "Your Clan" button), click **"Find Clan"**
2. List of public clans displays with details:
   - Clan name and tag
   - Member count (e.g., "15/30 members")
   - Clan rating/points

[Screenshot: Newly created clan page showing clan details, your name as leader, and empty statistics]

Figure 30: New Clan Page - Clan leader view after creation

- Description summary
- Minimum rating requirement

3. Use search/filter options to find specific clans:

- Search by clan name
- Filter by rating requirement
- Sort by clan points or member count

[Screenshot: Clan browser showing list of available clans with details and "Request to Join" buttons]

Figure 31: Clan Browser - List of available clans to join

Step 2: Select and Request to Join

1. Click on a clan to view full details

2. Clan detail page shows:
    - Complete clan information
    - Member list with usernames and ratings
    - Recent battle history
    - Clan statistics
3. If you meet requirements, **"Request to Join"** button is enabled
4. Click **"Request to Join"** button
5. Confirmation message: "Join request sent to [Clan Name]"

[Screenshot: Clan detail page with "Request to Join" button highlighted]

Figure 32: Clan Details - Request to join a clan

Step 3: Wait for Approval

1. Your join request is sent to the clan leader
2. Status shows: "Pending Approval"
3. Clan leader receives notification of your request
4. Leader can view your profile, rating, and statistics
5. Leader approves or rejects your request

Step 4: Join Request Accepted

1. When leader approves, you receive notification: "You have been accepted to [Clan Name]!"
2. Click notification or navigate to "Your Clan"

3. Your clan page now displays:
    - Clan information
    - Full member list (including yourself)
    - Clan chat/communication features
    - Clan battle schedule
4. You can now participate in clan battles when selected by leader

[Screenshot: Clan page showing member list with your name included, clan stats, and chat]

Figure 33: Your Clan Page - Member view after joining

Clan Leader Functions:
If you are a clan leader, additional management options are available:

1. **Managing Join Requests**:
    - Navigate to **"ClashConnect"** or check notifications
    - View pending join requests with applicant details
    - Click **"Accept"** to approve or **"Reject"** to deny
2. **Managing Members**:
    - View full member list in clan page
    - Option to **"Remove Member"** (kick from clan)
    - Promote members to co-leader (if feature available)
3. **Initiating Clan Battles**:
    - Select warriors for upcoming battles

- Start matchmaking for clan wars
- (Detailed in next subsection)

4. **Edit Clan Settings**:
   - Update description, logo, or settings
   - Change clan type (public/private)
   - Adjust member limit or rating requirements

[Screenshot: Clan leader view showing join requests panel with Accept/Reject buttons and member management options]

Figure 34: Clan Leader Interface - Managing requests and members

Leaving a Clan:

1. From your clan page, locate **"Leave Clan"** button (typically at bottom or in settings)
2. Click **"Leave Clan"**
3. Confirmation dialog: "Are you sure you want to leave [Clan Name]?"
4. Click **"Confirm"** to leave
5. Success message: "You have left the clan"
6. Redirected to clan browsing menu
7. Note: If you are the leader, you must transfer leadership or disband clan first

Input Requirements:

- **Creating Clan**: Clan name (unique), tag, description, type, member limit, minimum rating

- **Joining Clan**: Must meet clan's rating requirement (if any), cannot be in another clan
- **Leader Actions**: Select members for battles, approve/reject requests

Output and Results:

- **Clan Creation Output**: New clan created with you as leader, clan visible in browser
- **Join Request Output**: Request sent to leader, pending status
- **Membership Output**: Access to clan page, chat, battles, statistics
- **Management Output**: Notifications for join requests, member updates

5.3.2 Clan Battles

Purpose: Participate in team-based coding competitions where clans compete against each other. Selected clan members solve problems to earn points for their clan, with cumulative scores determining the winning team.

Step-by-Step Usage (Clan Leader):

Step 1: Access Clan Battle Interface

1. From Main Dashboard, click **"Clan Battle"** button on left navigation
2. As clan leader, you are taken to the warrior selection page
3. Page displays list of all clan members with their ratings and availability

[Screenshot: Clan Battle button highlighted on dashboard with notification badge]

Figure 35: Main Dashboard - Clan Battle access

Step 2: Select Warriors

1. Choose members to participate in the clan battle:
   - Typically 3-5 warriors per battle (varies by system configuration)
   - Click on member names to select/deselect
   - Selected members are highlighted
   - Member ratings displayed to help make strategic selections
2. Review selected warriors
3. Click **"Confirm Selection"** or **"Ready for Battle"** button

[Screenshot: Warrior selection interface showing clan member list with checkboxes and Confirm button highlighted]

Figure 36: Clan Battle - Warrior selection by leader

Step 3: Initiate Matchmaking

1. After confirming warrior selection, click **"Find Opponent Clan"** or **"Start Matchmaking"**
2. System searches for another clan ready for battle with similar:
   - Clan rating
   - Number of warriors
   - Battle tier
3. Searching status displays: "Searching for opponent clan..."
4. Matchmaking may take 30 seconds to several minutes depending on availability
5. You can cancel matchmaking by clicking **"Cancel Search"**

Step 4: Opponent Clan Found - Battle Setup

[Screenshot: Matchmaking screen showing "Searching for opponent clan" with loading animation]

Figure 37: Clan Battle - Matchmaking in progress

1. When match found, notification appears: "Opponent Clan Found!"
2. Battle details displayed:
   - Opponent clan name and tag
   - Opponent warrior list with ratings
   - Battle format (e.g., "Best of 5 problems", "30 minutes duration")
   - Problem difficulty range
3. Countdown timer (5-10 seconds) before battle begins
4. All selected warriors from both clans receive notification to enter battle

[Screenshot: Opponent clan found screen showing clan names, warrior lists, and countdown timer]

Figure 38: Clan Battle - Opponent found and battle initialization

Step 5: Clan Battle Arena (Warrior Perspective)
For selected warriors (regular members):

1. Notification received: "You have been selected for Clan Battle!"
2. Click notification or navigate to **"Clan Battle"** button
3. Battle arena loads showing:
   - **Top Bar**: Your clan vs opponent clan, current score (e.g., "3-2"), battle timer
   - **Problem Panel**: List of problems (e.g., Problem 1, Problem 2, Problem 3)
   - **Teammate Status**: Which teammates are solving which problems
   - **Opponent Status**: How many problems opponent clan has solved
4. Select a problem to solve by clicking on it
5. Problem statement and code editor load (similar to PvP battle interface)

[Screenshot: Clan battle arena showing clan names, scores, problem list, and teammate status]

Figure 39: Clan Battle Arena - Warrior view with problem selection

Step 6: Solving Problems as Team

1. Warriors can solve any available problem:
   - Multiple warriors can attempt the same problem
   - Or distribute problems strategically (coordination via clan chat)
2. Solve problems using standard code editor:
   - Write code

- Test locally
- Submit solution

3. When any warrior gets Accepted verdict:
   - Point awarded to your clan for that problem
   - Problem marked as solved (checkmark appears)
   - Notification sent to all clan members: "[Warrior Name] solved Problem X!"
   - Clan score increments
4. Continue solving remaining problems
5. Real-time score updates: "Your Clan: 4 - Opponent Clan: 3"

[Screenshot: Problem being solved in clan battle, showing code editor and teammate solving another problem in status panel]

Figure 40: Clan Battle - Warriors solving problems collaboratively

Step 7: Clan Battle Conclusion

1. Battle ends when:
   - Time limit reached (e.g., 30 minutes)
   - One clan solves all problems
   - Predetermined winning score reached
2. Result screen displays:
   - **Winner Announcement**: "Victory!" or "Defeat!"
   - **Final Score**: Your Clan vs Opponent Clan (e.g., "5-4")
   - **Warrior Contributions**: Which warriors solved which problems

- **Clan Points Earned**: Points added to clan rating
- **Individual XP**: Each warrior earns XP based on contribution

3. Battle summary added to clan battle history
4. Clan leaderboards updated with new points
5. Click **"Return to Clan Page"** or **"Back to Dashboard"**

[Screenshot: Clan battle result screen showing Victory, final score, warrior contributions, and points earned]

Figure 41: Clan Battle Result - Victory screen with statistics

Non-Selected Members During Battle:
If you are a clan member but not selected for current battle:

1. Clicking "Clan Battle" button shows: "No Battle Ongoing" or "Battle in progress - You are not participating"
2. You can spectate (if feature available):
   - View live scores
   - See which warriors are solving problems
   - Cannot interact with battle
3. Wait for next battle to potentially be selected

Input Requirements:

- **Leader Input**: Select 3-5 warriors from clan members, initiate matchmaking
- **Warrior Input**: Accept battle notification, select problems to solve, submit code solutions

- **Team Strategy**: Optional coordination via clan chat

Output and Results:

- **Matchmaking Output**: Opponent clan found with details
- **Battle Output**: Real-time battle arena with problems, scores, teammate status
- **Verdicts**: Individual submission results for each warrior
- **Battle Result**: Win/Loss/Draw, final score, clan points earned/lost, individual XP rewards
- **Clan Statistics**: Updated clan rating, battle history, win/loss record

Special Features:

- **Team Collaboration**: Multiple warriors working together
- **Strategic Selection**: Leaders choose best warriors for battles
- **Real-Time Coordination**: Live updates on teammate progress
- **Clan Ranking**: Compete for top clan positions
- **Collective Victory**: Team success benefits all members

5.4 Social Features

5.4.1 ClashConnect - Friend and Clan Management Hub

Purpose: ClashConnect serves as the central social hub for managing friend connections, viewing and responding to clan join requests (for leaders), sending battle invitations, and browsing other players' profiles.

Step-by-Step Usage:

Step 1: Access ClashConnect

1. From Main Dashboard, click **"ClashConnect"** button on the left navigation menu
2. If you have pending notifications (friend requests, clan join requests), a red notification badge appears on the button
3. ClashConnect interface opens showing multiple tabs or sections

Step 2: Browse Friend List

1. ClashConnect displays your current friend list showing:
    - Friend's username

[Screenshot: Main Dashboard with ClashConnect button highlighted, showing notification badge]

Figure 42: Main Dashboard - ClashConnect access with notifications

- Current online status (green dot = online, gray = offline)
- Rating and level
- Quick action buttons: View Profile, Challenge to Battle, Remove Friend

2. Click on a friend to view detailed profile

3. Use **"Challenge to Battle"** button to send Local Battle invitation

[Screenshot: ClashConnect friend list showing usernames, online status, and action buttons]

Figure 43: ClashConnect - Friend list with online status

Step 3: Search for New Friends

1. Navigate to **"Find Friends"** or **"Search Users"** tab/section

2. Enter username in search field

3. Search results display matching users with:
    - Username
    - Rating and level
    - Clan affiliation (if any)
    - Current relationship status (Not Friends, Request Sent, Request Received)
4. Click **"Add Friend"** button next to desired user
5. Confirmation: "Friend request sent to [username]"

[Screenshot: Find Friends search interface with search bar and "Add Friend" buttons highlighted]

Figure 44: ClashConnect - Search for new friends

Step 4: Manage Friend Requests

1. Navigate to **"Pending Requests"** or **"Notifications"** section
2. View incoming friend requests showing:
    - Requester's username
    - Rating and level
    - Time of request
3. For each request:
    - Click **"Accept"** to add as friend → Success: "[Username] is now your friend"
    - Click **"Reject"** to decline → Request removed
    - Click username to view full profile before deciding
4. Accepted friends immediately appear in your friend list

[Screenshot: Pending friend requests with Accept and Reject buttons highlighted]

Figure 45: ClashConnect - Managing incoming friend requests

Step 5: Clan Join Requests (Clan Leaders Only)

If you are a clan leader, ClashConnect also displays clan join requests:

1. Navigate to **"Clan Requests"** tab (only visible to leaders)
2. View users who requested to join your clan:
   - Applicant username
   - Rating and level
   - Current clan status (not in clan / left previous clan)
   - Application time
3. Click on applicant to view detailed profile and statistics
4. Decision options:
   - **"Accept"**: User joins your clan immediately, receives notification
   - **"Reject"**: Request denied, user can apply again or join other clans
5. Clan member list updates automatically when requests are accepted

Step 6: View Player Profiles

1. Click on any username (from friend list, search results, or leaderboards)
2. Player profile page displays:
   - Username and avatar
   - Rating, level, and XP progress

[Screenshot: Clan join requests panel showing applicant details with Accept/Reject buttons (clan leader view)]

Figure 46: ClashConnect - Clan join requests (leader view)

- Clan affiliation and tag
- Battle statistics: Win/Loss record, total battles, win rate percentage
- Recent battle history
- Problems solved count
- Join date

3. Action buttons available:
   - **"Add Friend"** (if not friends)
   - **"Challenge to Battle"** (if friends or available)
   - **"View Clan"** (if player is in a clan)
   - **"Remove Friend"** (if already friends)
4. Click **"Back"** to return to ClashConnect

Input Requirements:

- **Friend Search**: Username or partial username
- **Friend Request**: Target user must exist and not already be friend
- **Clan Request Management (Leaders)**: Decision to accept or reject applicants

Output and Results:

- **Friend List Output**: Display of all friends with online status
- **Search Output**: List of users matching search query

[Screenshot: Player profile page showing stats, rating, clan, battle history, and action buttons]

Figure 47: Player Profile - Detailed statistics and social actions

- **Request Output**: Notifications for pending friend/clan requests
- **Profile Output**: Detailed player information and statistics
- **Relationship Updates**: Friend added/removed confirmations

Special Features:

- **Centralized Hub**: All social features in one interface
- **Real-Time Notifications**: Instant updates for new requests
- **Online Status**: See which friends are currently active
- **Quick Challenges**: Send battle invitations directly from friend list
- **Clan Management Integration**: Leaders manage both friends and clan members

5.5 Leaderboards

Purpose: View global and clan rankings to track top performers, compare your standing with other players, and monitor clan competition.

Step-by-Step Usage:

Step 1: Access Leaderboards

1. From Main Dashboard, locate the **"Leaderboards"** button (may be in navigation menu or separate section)
2. Click **"Leaderboards"** to open ranking interface
3. Leaderboard page loads showing ranking tabs

[Screenshot: Navigation menu with Leaderboards button highlighted]

Figure 48: Access Leaderboards from navigation

Step 2: Global Player Leaderboard

1. Default view shows **"Global Player Rankings"**
2. Leaderboard displays ranked list of players:
   - **Rank**: Position number (1st, 2nd, 3rd, etc.)
   - **Username**: Player name (clickable to view profile)
   - **Rating**: Current ELO rating
   - **Level**: Player level
   - **XP**: Total experience points
   - **Win/Loss**: Battle record
   - **Clan Tag**: Clan affiliation (if any)
3. Top 3 players highlighted with special badges (gold, silver, bronze)
4. Your own rank is highlighted in different color for easy identification
5. Pagination: Navigate through pages to see more players

Step 3: Clan Leaderboard

1. Click on **"Clan Rankings"** tab
2. Leaderboard switches to clan rankings showing:
   - **Rank**: Clan position
   - **Clan Name and Tag**: Clan identifier
   - **Clan Points**: Total clan rating/points

[Screenshot: Global leaderboard showing ranked player list with columns for rank, username, rating, level, and win/loss]

Figure 49: Global Player Leaderboard - Top ranked players

- **Members**: Total member count
- **Battles**: Total clan battles fought
- **Win Rate**: Percentage of battles won
- **Leader**: Clan leader username

3. Top clans highlighted with special styling
4. Your clan (if you're in one) is highlighted
5. Click on clan name to view clan details page

[Screenshot: Clan leaderboard showing ranked clans with points, members, and win rates]

Figure 50: Clan Leaderboard - Top ranked clans

Step 4: Filter and Sort Options

1. Use filter/sort options to customize leaderboard view:
   - **Time Period**: All-Time, Monthly, Weekly rankings
   - **Category**: Rating, XP, Battles Won, Problems Solved
   - **Region**: Global, or by specific region (if implemented)
2. Select filter/sort criteria
3. Leaderboard updates dynamically to show filtered results

Step 5: View Detailed Player/Clan Profile

1. Click on any player username or clan name in the leaderboard
2. Redirected to detailed profile/clan page
3. View complete statistics and battle history
4. Option to add friend, challenge, or join clan (depending on context)
5. Click **"Back to Leaderboard"** to return

Input Requirements:

- **No Direct Input**: Leaderboards are view-only, automatically populated
- **Optional Filters**: Select time period or category for custom rankings

Output and Results:

- **Player Rankings**: Sorted list of players by rating, XP, or other criteria
- **Clan Rankings**: Sorted list of clans by points and performance
- **Your Position**: Highlighted rank showing your standing
- **Statistics**: Win rates, battle counts, and performance metrics
- **Real-Time Updates**: Rankings update after each battle completion

Special Features:

- **Live Rankings**: Real-time updates as battles conclude
- **Multiple Categories**: Different ranking criteria available
- **Top Player Recognition**: Special badges for top 3 positions
- **Personal Tracking**: Easy identification of your own rank
- **Competitive Motivation**: Track progress toward top positions

5.6 Profile and Statistics

Purpose: View and manage your personal profile, track your coding progress, monitor statistics, and update account settings including Codeforces handle.

Step-by-Step Usage:

Step 1: Access Your Profile

1. From Main Dashboard, click on your **username** or **profile icon** (typically in top-right corner or navigation menu)
2. Dropdown menu appears with options including **"View Profile"** or **"My Profile"**
3. Click **"View Profile"**
4. Your profile page loads

[Screenshot: User menu dropdown with "View Profile" option highlighted]

Figure 51: Access Profile from user menu

Step 2: Profile Overview

Your profile displays comprehensive information:

- **Personal Information**:
  - Username
  - Avatar/profile picture (if available)
  - Join date
  - Email address (partially hidden for privacy)
  - Linked Codeforces handle
- **Progression Statistics**:

  - Current Level (e.g., Level 15)
  - XP Bar showing progress to next level (e.g., "2,450 / 3,000 XP")
  - Total XP earned

- **Rating and Ranking**:
  - Current ELO rating (e.g., 1,580)
  - Global rank (e.g., "#247 Globally")
  - Rating progress graph (optional visual chart)

- **Battle Statistics**:
  - Total battles fought
  - Wins / Losses / Draws
  - Win rate percentage (e.g., "65% win rate")
  - Average battle time

- **Problem Solving Stats**:
  - Total problems solved
  - Problems by difficulty (Easy: X, Medium: Y, Hard: Z)
  - Favorite problem tags

- **Clan Information**:
  - Current clan name and tag (if in a clan)
  - Role in clan (Leader / Member)
  - Clan contribution points

Step 3: View Battle History

1. Scroll down to **"Recent Battles"** section in profile
2. Or click **"View Full Battle History"** button
3. Battle history displays list of past battles:
   - Date and time of battle
   - Battle type (Global / Local / Clan)
   - Opponent username (or clan name for clan battles)
   - Problem solved (with link to Codeforces problem)

[Screenshot: Complete profile page showing all statistics sections (personal info, rating, battles, problems, clan)]

Figure 52: User Profile - Complete statistics overview

- Result (Win / Loss / Draw)
- Rating change (e.g., "+12", "-8")
- Time taken

4. Click on individual battle to see detailed battle summary

5. Pagination available for viewing older battles

[Screenshot: Battle history section showing list of past battles with results and rating changes]

Figure 53: Battle History - Record of all past battles

Step 4: Edit Profile Settings

1. Click **"Edit Profile"** or **"Settings"** button

2. Profile settings page opens with editable fields:

- **Codeforces Handle**: Update linked Codeforces username
  - Enter new handle
  - System validates handle exists on Codeforces
  - Note: Handle changes may be rate-limited (e.g., once per month)
- **Avatar**: Upload profile picture (if feature available)
- **Bio**: Add personal description (optional)
- **Notification Preferences**: Enable/disable email notifications for:
  - Friend requests
  - Battle invitations
  - Clan announcements
- **Privacy Settings**: Control who can:
  - Send you friend requests
  - Challenge you to battles
  - View your battle history

3. Make desired changes
4. Click **"Save Changes"** button
5. Success notification: "Profile updated successfully"

[Screenshot: Profile settings page with Codeforces handle field, notification toggles, and Save button highlighted]

Figure 54: Profile Settings - Edit personal information and preferences

Step 5: View Progress Graphs (Optional)

If analytics are available:

1. Navigate to **"Statistics"** or **"Analytics"** tab in profile

2. View visual representations:
   - **Rating Progress Graph**: Line chart showing rating changes over time
   - **Battle Activity**: Calendar heatmap or bar chart of battles per day/week
   - **Problem Difficulty Distribution**: Pie chart of solved problems by difficulty
   - **Win Rate Trend**: Graph showing win rate over time
3. Hover over graph elements for detailed tooltips
4. Use date range selector to filter time periods

[Screenshot: Statistics page with rating graph and problem distribution charts]

Figure 55: Profile Statistics - Visual analytics and progress tracking

Input Requirements:

- **Profile Access**: No input required (automatic based on logged-in user)
- **Settings Update**: New Codeforces handle (must be valid), avatar image (if uploading), preference selections

Output and Results:

- **Profile Display**: Complete statistics, ratings, battle history
- **Progress Tracking**: XP bar, level, rating changes
- **Battle Record**: Detailed history of all battles
- **Settings Confirmation**: Success messages for updates
- **Analytics**: Visual graphs of progress (if available)

Special Features:

- **Comprehensive Stats**: All-in-one view of your coding journey
- **Progress Tracking**: Visual XP bar and level progression
- **Rating History**: Track rating changes after each battle
- **Codeforces Integration**: Update linked handle for submissions
- **Privacy Controls**: Manage who can interact with you

5.7 Code Editor Features

Purpose: The integrated code editor provides a comprehensive coding environment within the browser, supporting multiple programming languages, syntax highlighting, local testing, and seamless submission to Codeforces judge.

Key Features Overview:

The code editor is accessible in all coding contexts (Global Battles, Local Battles, Clan Battles, Practice Gym) and provides consistent functionality across all modes.

Step-by-Step Usage:

Step 1: Editor Interface

When entering any battle or practice session, the code editor appears with:

- **Editing Area**:
  - Large text area for writing code
  - Line numbers displayed on left margin
  - Syntax highlighting based on selected language
  - Auto-indentation support
  - Code folding for functions/blocks (if available)
- **Top Toolbar**:
  - Language selector dropdown
  - Font size adjustment (if available)
  - Theme toggle (light/dark mode, if available)
- **Bottom Action Bar**:
  - **"Run Code"** button (local testing)
  - **"Submit"** button (official submission)
  - **"Reset"** button (clear code)

Step 2: Select Programming Language

[Screenshot: Code editor interface showing editing area, language dropdown, line numbers, and action buttons]

Figure 56: Code Editor - Main interface components

1. Click on the **language dropdown menu** (typically shows default like "C++")
2. Available languages include:
   - C++ (multiple versions: C++11, C++14, C++17, C++20)
   - Java (Java 8, Java 11, Java 17)
   - Python (Python 2, Python 3)
   - JavaScript (Node.js)
   - C (C99, C11)
   - Others: Go, Rust, Kotlin (depending on Codeforces support)
3. Select your preferred language
4. Editor syntax highlighting automatically updates to match selected language
5. Template code may auto-populate (e.g., `int main()` for C++, `public class Main` for Java)

Step 3: Write Code

1. Click in the editing area and start typing your solution
2. Editor features while typing:
   - **Syntax Highlighting**: Keywords, strings, comments colored for readability
   - **Auto-Indentation**: Automatic indentation when pressing Enter
   - **Bracket Matching**: Matching brackets highlighted when cursor adjacent

[Screenshot: Language dropdown expanded showing list of available programming languages]

Figure 57: Language Selection - Choose programming language

- **Tab Support**: Tab key inserts spaces (typically 2 or 4 spaces)
- **Undo/Redo**: Ctrl+Z / Ctrl+Y for undo and redo
- **Copy/Paste**: Standard clipboard operations (Ctrl+C, Ctrl+V)
- **Find/Replace**: Ctrl+F to search within code (if available)

3. Write your complete solution
4. Use line numbers for reference and debugging

[Screenshot: Code editor with sample code written, showing syntax highlighting and line numbers]

Figure 58: Code Editor - Writing code with syntax highlighting

Step 4: Local Testing with Piston API

1. After writing code, click **"Run Code"** or **"Test"** button

2. Test input panel appears:
   - **Input Field**: Text area to enter test input data
   - Sample test cases from problem may be pre-filled
   - You can modify or enter custom test cases
3. Enter input data matching problem's input format
4. Click **"Run"** or **"Execute"** button
5. Code executes locally using Piston API (remote code execution service)
6. Wait for execution (typically 1-3 seconds)
7. Test results display:
   - **Your Output**: What your code produced
   - **Expected Output**: Sample output from problem (if available)
   - **Execution Time**: Time taken in milliseconds
   - **Memory Used**: Memory consumption (if available)
   - **Status**: Success (ran without errors) or Error (runtime/compilation error)
8. If compilation error or runtime error:
   - Error message displays with details
   - Line number of error (if available)
   - Fix code and test again
9. Run multiple test cases to verify correctness

Step 5: Submit to Codeforces Judge

1. Once confident code is correct, click **"Submit"** button
2. Confirmation dialog may appear: "Submit code to Codeforces judge?"
3. Click **"Confirm"**
4. Submission process begins:
   - Code sent to Codeforces servers
   - Your linked Codeforces handle used for submission
   - Selected language and problem ID included in submission

[Screenshot: Test panel showing input field, Run button, and output results comparing expected vs actual output]

Figure 59: Local Testing - Test code with custom inputs

5. Submission status updates in real-time:
   - "Submitting..." (sending code to Codeforces)
   - "In Queue" (waiting for judge)
   - "Running on test 1", "Running on test 2", ... (executing against test cases)
6. Final verdict appears (typically within 5-15 seconds):
   - **Accepted (AC)**: All test cases passed (green color)
   - **Wrong Answer (WA)**: Failed on test case X (red color)
   - **Time Limit Exceeded (TLE)**: Code too slow on test X (yellow/orange)
   - **Runtime Error (RE)**: Code crashed on test X (red color)
   - **Memory Limit Exceeded (MLE)**: Used too much memory
   - **Compilation Error (CE)**: Code doesn't compile
7. Detailed verdict information shown:
   - Test case number where error occurred (if failed)
   - Execution time for each test (if available)
   - Verdict message from Codeforces

Step 6: Handle Verdicts
If Accepted:

- Success notification displays
- In battles: You win the battle (or contribute point to clan)

[Screenshot: Submission in progress showing "Running on test 5" status with progress indicator]

Figure 60: Code Submission - Real-time verdict tracking

- In practice: Problem marked as solved, XP awarded
- Option to view full submission details on Codeforces

If Not Accepted:

- Error details displayed
- You can modify code and resubmit
- Unlimited submissions allowed (in practice and during battle time)
- Use local testing to debug before resubmitting

[Screenshot: Verdict display showing "Accepted" in green with checkmark, or "Wrong Answer on test 5" in red]

Figure 61: Submission Verdict - Success or error display

Step 7: Code Management

1. **Reset Code**: Click "Reset" button to clear editor and start fresh
2. **Save Draft**: Some implementations auto-save code in browser localStorage
3. **Copy Code**: Select all code (Ctrl+A) and copy (Ctrl+C) to save externally
4. **Load Template**: Reset may load language-specific template code

Input Requirements:

- **Language Selection**: Choose from available languages
- **Code Input**: Valid syntax for selected language
- **Test Input**: Sample or custom test data matching problem format
- **Submission Requirement**: Codeforces handle must be linked

Output and Results:

- **Editor Output**: Syntax-highlighted code with line numbers
- **Local Test Output**: Execution results, output, runtime, errors
- **Submission Output**: Real-time status updates, final verdict from Codeforces
- **Error Feedback**: Compilation errors, runtime errors, wrong answer details

Supported Languages Summary:

Language	**Versions/Notes**
C++	C++11, C++14, C++17, C++20 (most popular for competitive programming)
Java	Java 8, Java 11, Java 17 (class name must be Main)
Python	Python 2.7, Python 3.x (slower execution, suitable for easier problems)
C	C99, C11 (lower-level alternative to C++)
JavaScript	Node.js runtime (less common for competitive programming)
Others	Go, Rust, Kotlin (availability depends on Codeforces support)

Special Features:

- **Multi-Language Support**: Switch languages anytime before submission
- **Syntax Highlighting**: Improved code readability

- **Local Testing**: Test before submitting (no submission penalty)
- **Real-Time Verdicts**: Live updates during Codeforces judging
- **Unlimited Submissions**: No limit on resubmissions
- **Codeforces Integration**: Official verdict from competitive programming platform

5.8 Rating and Progression System

Purpose: Track your skill development through an ELO-based rating system and level-based progression with experience points (XP). The system provides competitive rankings and motivational milestones.

How the System Works:

ELO-Based Rating System:

- Every user starts with an initial rating (typically 1200-1500)
- Rating changes after each battle based on:
  - Battle outcome (Win/Loss/Draw)
  - Opponent's rating (beating higher-rated opponents earns more points)
  - Expected outcome probability (ELO formula)
- Rating change examples:
  - Beat opponent with similar rating: +12 to +15 points
  - Beat opponent with much higher rating: +25 to +40 points
  - Lose to similar rating opponent: -10 to -13 points
  - Lose to much lower rating opponent: -30 to -50 points
- Rating is used for:
  - Matchmaking in Global Battles (±200 rating range)
  - Global leaderboard rankings
  - Skill level indication

Experience Points (XP) and Levels:

- XP is earned through various activities:
  - Winning battles: 100-200 XP (varies by opponent rating)
  - Losing battles: 20-50 XP (participation reward)

  - Solving practice problems: 30-80 XP (based on difficulty)
  - Contributing to clan battle wins: 80-150 XP
  - Daily login bonus: 10 XP (if implemented)

- Levels increase as XP accumulates:
  - Level 1: 0 XP required (starting level)
  - Level 2: 100 XP required
  - Level 3: 300 XP required
  - Level 10: 2,000 XP required
  - Higher levels require exponentially more XP
- Levels are purely cosmetic/motivational (do not affect matchmaking)
- XP bar displayed in profile and dashboard shows progress to next level

Viewing Rating and XP:
Step 1: Dashboard XP Bar

1. After logging in, Main Dashboard displays XP bar at top of navigation menu
2. XP bar shows:
   - Current level (e.g., "Level 12")
   - Current XP / Required XP for next level (e.g., "1,450 / 2,500 XP")
   - Visual progress bar (filled proportionally)
   - Username
3. Progress bar fills as you earn XP
4. Level-up notification appears when reaching XP threshold: "Level Up! You are now Level 13"

Step 2: View Rating in Profile

1. Navigate to your profile (click username → "View Profile")
2. Profile displays current rating prominently:
   - Large number showing current rating (e.g., "1,580")
   - Rating change since last battle (e.g., "+12" in green, or "-8" in red)
   - Rating tier/rank (if tiers are implemented: Bronze, Silver, Gold, etc.)

[Screenshot: Dashboard with XP bar highlighted showing level, XP progress, and username]

Figure 62: XP Bar - Level and experience point tracking

[Screenshot: Profile page with rating prominently displayed and rating change indicator]

Figure 63: Profile Rating Display - Current rating and global rank

- Global rank based on rating (e.g., "#247 Globally")

3. Rating history graph (if available) shows rating changes over time

Step 3: Rating Changes After Battles

1. After each battle concludes, battle result screen displays:
   - **"Rating Change"**: Exact points gained or lost (e.g., "+15 Rating")
   - Color coding: Green for gain, red for loss
   - New current rating (e.g., "Your rating: 1,595")
2. XP earned also displayed: "XP Earned: +120"
3. Updated statistics reflected immediately in profile

[Screenshot: Battle result screen highlighting rating change "+15" and XP earned "120 XP"]

Figure 64: Battle Result - Rating and XP changes

Step 4: Leaderboard Rankings

1. Access Leaderboards to see how your rating compares globally
2. Your rank is determined solely by rating
3. Higher rating = higher position on leaderboard
4. Compete to reach top positions and top tiers

Rating Tiers/Divisions (If Implemented):

Tier	Rating Range	Description
Bronze	0-1199	Beginner tier, learning fundamentals
Silver	1200-1599	Intermediate tier, solid understanding
Gold	1600-1999	Advanced tier, strong problem-solving skills
Platinum	2000-2399	Expert tier, competitive programming expertise
Diamond	2400+	Master tier, top competitive programmers

XP Earning Activities Summary:

Activity	XP Earned	Notes
Win Global Battle	100-200 XP	More XP for beating higher-rated opponents
Lose Global Battle	20-50 XP	Participation reward
Win Local Battle	80-150 XP	Depends on implementation

Activity	XP Earned	Notes
Win Clan Battle	150-250 XP	Team contribution bonus
Solve Practice Problem	30-80 XP	Based on problem difficulty rating
Daily Login	10 XP	If daily bonus implemented
First Win of the Day	50 XP bonus	If daily streak implemented

Input Requirements:

- **No Direct Input**: Rating and XP calculated automatically based on activity
- **Participation**: Engage in battles and practice to earn XP and rating

Output and Results:

- **Rating Display**: Current rating, changes, global rank
- **XP Progress**: Visual bar, level, XP to next level
- **Level-Up Notifications**: Alerts when reaching new level
- **Statistics**: Rating history, XP earned over time
- **Tier Badges**: Visual indicators of skill tier (if implemented)

Special Features:

- **Fair ELO System**: Rating reflects true skill level through balanced calculations
- **Motivational Progression**: XP and levels provide constant progress feedback
- **Competitive Ranking**: Global leaderboard based on rating
- **Visual Tracking**: XP bar and rating graphs show improvement
- **Multiple Rewards**: Earn XP from battles, practice, and clan activities

6 Navigation Guide

This section provides a comprehensive guide to navigating CodeClash Arena's interface. After logging in, you will be directed to the Main Dashboard, which serves as the central hub for accessing all platform features.

6.1 Main Dashboard Overview

The Main Dashboard features an animated background with coding-themed particles and a vertical menu bar on the left side. At the top of the menu, you will see your XP bar displaying your current level, experience points, and username. Below the XP bar are five main navigation buttons and a logout option.

6.2 Navigation Options

6.2.1 Your Clan

Access your clan management interface and view clan details. If you are not currently in a clan, clicking this button opens the clan browsing menu where you can search for clans to join or create your own clan. If you are already a clan member, this opens your clan's information page showing member list, clan statistics, total points, and war history.

6.2.2 ClashConnect

Open the social hub to manage your connections and invitations. This feature allows you to search for other players, send and accept friend requests, view incoming clan join requests (if you are a clan leader), and manage your friend list. A notification badge appears on this button when you have pending friend requests or clan join requests.

6.2.3 Clan Battle

Navigate to the clan battle interface for team-based competitions. If you are a clan leader, clicking this button takes you to the team selection page where you can choose warriors and initiate matchmaking for clan battles. If you are a regular clan member and a battle is ongoing, you will be directed to the battle arena; otherwise, you will see a "No Battle Ongoing" message. A notification badge appears when there is an active clan battle.

6.2.4 Attack

Launch the PvP battle menu to compete in 1v1 coding challenges. This button opens an overlay menu with multiple battle mode options: Global Battle Mode (matchmaking with players of similar rating), Local Battle Mode (challenge specific friends), and Practice Gym (solo practice without competition). Select your preferred mode to begin competitive coding.

6.2.5 Battle History

View your complete battle record and performance statistics. This section displays all your past 1v1 battles and clan battles, showing opponents faced, problems solved, battle outcomes (win/loss), time taken, and rating changes. Use this feature to analyze your performance trends and track your improvement over time.

6.2.6 Log Out

Securely end your current session and return to the homepage. Clicking this button clears your authentication tokens and logs you out of the platform. Always use this

option when accessing CodeClash Arena from shared or public computers to protect your account security.

7 Error Handling and Messages

CodeClash Arena displays color-coded alerts to communicate system status: green alerts for successful actions (e.g., "Sign in successful!") and red alerts for errors. Alerts disappear automatically after a few seconds or can be manually closed.

7.1 Authentication Errors

Invalid Login Credentials

Error Message: "Invalid login credentials" or "Invalid email or password"

Solutions: Verify email and password are correct; check Caps Lock is off; ensure no extra spaces; use correct email (not Codeforces handle); verify email has been confirmed after signup.

Email Not Confirmed/Verified

Error Message: "Email not confirmed" or "Error fetching user data"

Solutions: Check inbox and spam folder for verification email from Supabase; look for subject "Confirm Your Email"; click verification link; return to login page after verification; contact support if link expired (24 hours).

Account Locked

Error Message: "Too many login attempts"

Solutions: Wait 15-30 minutes before retrying; use password reset if needed; avoid multiple attempts during lockout.

Email Already Registered

Error Message: "User already registered" or "Email already exists"

Solutions: Use login page if account exists; use password reset if password forgotten; try different email for new account.

Invalid Codeforces Handle

Error Message: Field validation error or "Invalid handle"

Solutions: Verify handle at `https://codeforces.com/profile/your-handle`; ensure exact spelling and capitalization; create Codeforces account first if needed.

Weak Password

Error Message: "Password must be at least 8 characters"

Solutions: Use minimum 8 characters with mix of uppercase, lowercase, numbers, and special characters; avoid common passwords.

7.2 Network and Connectivity Errors

Network Error: "Network error" or "Failed to fetch"

Solutions: Check internet connection; refresh page (F5); verify firewall settings; disable VPN temporarily; clear browser cache; try different browser.

Timeout Error: "Request timeout" or "Connection timeout"

Solutions: Ensure stable connection (5+ Mbps for battles); close bandwidth-consuming applications; switch to wired connection; try again after a moment.

7.3 Battle-Related Errors

Battle Disconnection: "Connection lost" or "Battle disconnected"

Note: Automatic forfeit if reconnection fails.

Prevention: Use wired Ethernet connection; ensure stable internet before battles; close unnecessary applications; avoid battles during network instability.

Submission Failed: "Submission failed" or "Error submitting code"

Solutions: Check internet connection; verify Codeforces handle is correctly linked; ensure Codeforces account is in good standing; retry after a few seconds; verify correct programming language selected.

7.4 Clan-Related Errors

Cannot Join Clan: "Already in a clan" or "Join request failed"

Solutions: Leave current clan before joining another; wait for pending requests to be processed; verify clan is accepting members and you meet requirements.

Cannot Start Clan Battle: "Insufficient permissions" or "Only clan leader can start battles"

Solutions: Contact clan leader to start battle; create your own clan for leadership permissions.

7.5 Common Error Messages Quick Reference

Error Message	**Quick Solution**
"Invalid login credentials"	Check email and password; ensure credentials correct; verify email confirmed
"Email not confirmed" / "Error fetching user data"	Check inbox for verification email; click verification link; check spam folder
"Network error"	Check internet; refresh page; verify firewall settings
"User already registered"	Use login instead; try different email; reset password if needed
"Too many login attempts"	Wait 15-30 minutes before retry

Error Message	Quick Solution
"Submission failed"	Check internet; verify Codeforces handle linked; retry submission
"Connection timeout"	Improve internet stability; use wired connection; close other apps
"Battle disconnected"	Ensure stable internet before battles; use wired connection
"Already in a clan"	Leave current clan first
"Only clan leader can start battles"	Contact clan leader or create own clan
"Password must be at least 8 characters"	Use longer password with mixed characters (uppercase, lowercase, numbers, symbols)

7.6 Getting Additional Help

If errors persist after trying solutions: take screenshot of error message; note exact steps leading to error; check browser console (F12 → Console) for details; contact platform support; verify system meets requirements in Section 2.